League Pro Database management system

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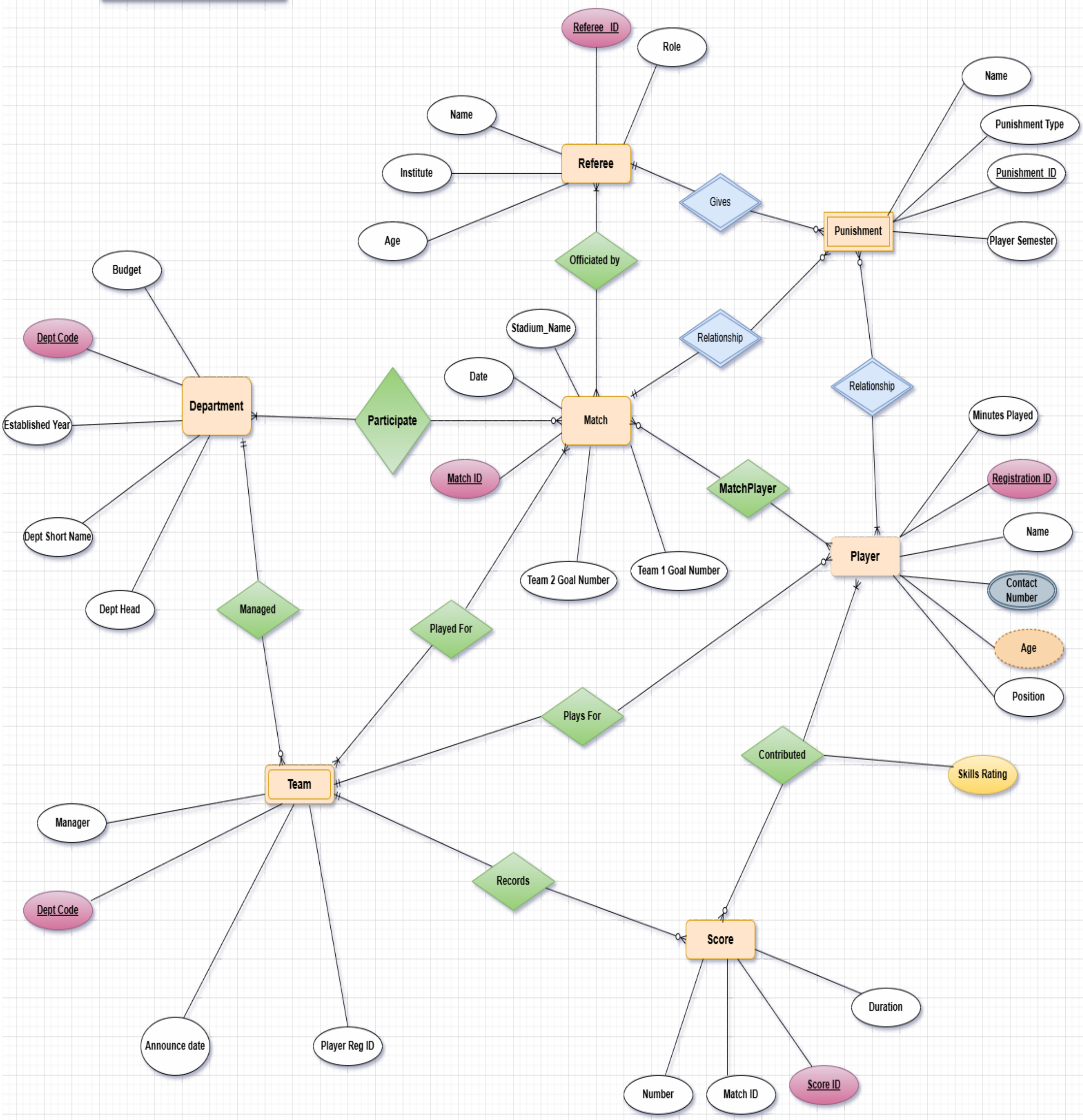

LeagueProDB: Football Tournament Database Report

1. Introduction

The **LeagueProDB** is a relational database designed for managing a football tournament. The schema ensures efficient storage, retrieval, and maintenance of data related to teams, players, referees, matches, scores, and punishments.

2. Database Schema Overview

The database consists of several entities and relationships:

2.1 Key Entities

- **Department**: Stores information about different football departments, their budgets, and heads.
- **Team**: Represents football teams, their managers, and related department codes.
- **Player**: Maintains player details such as name, age, contact information, position, and skills rating.
- **Match**: Records match details, including the stadium, date, and participating teams.
- **Score**: Tracks match scores, duration, and score types.
- **Referee**: Stores referee details, including their roles, institute, and age.
- **Punishment**: Manages disciplinary actions given to players during matches.

2.2 Relationships

- **Participate**: Links departments to matches they are involved in.
- **RefereeMatch**: Connects referees to the matches they officiate.
- **MatchPunishment**: Associates punishments with matches where they were given.
- **Have**: Links punishments to players who received them.
- **MatchPlayer**: Connects players with the matches they played in.
- **Contributed**: Tracks players who contributed to scores in matches.

3. Constraints and Data Integrity

- **Primary Keys (PKs)**: Unique identifiers are enforced on key attributes such as Team_ID, Player_ID, and Match_ID.
- **Foreign Keys (FKs)**: Relationships between entities are maintained using foreign keys to ensure referential integrity.
- **Check Constraints**: Various constraints are applied, such as:
  - Age constraints for players (10–50 years) and referees (18–65 years).
  - Skill ratings (0–10 range).
  - Budget and minutes played (must be non-negative).

4. Performance Optimization

- **Indexes**: Created on key attributes such as player names, match dates, and referee names to improve query performance.
- **Normalization**: The database follows a well-structured normalization approach to eliminate redundancy.

5. ER Diagram Analysis

The ER diagram visually represents the database schema, including:

- **Entities (Rectangles)**: Representing core database tables.
- **Attributes (Ellipses)**: Showing key fields.
- **Relationships (Diamonds)**: Connecting entities with appropriate cardinalities.

6. Use Cases and Applications

- **Tournament Management**: Efficiently track teams, players, and match details.
- **Player Performance Monitoring**: Maintain statistics on minutes played, scores contributed, and skills ratings.
- **Referee Assignments**: Manage referee allocations for different matches.
- **Disciplinary Actions**: Keep a record of punishments given to players.

7. Conclusion

The **LeagueProDB** schema provides a structured and optimized database solution for football tournament management. It ensures data integrity, efficient querying, and comprehensive tracking of all aspects of the tournament.